

New Keynesian Models: Study Notes

Based on Romer, *Advanced Macroeconomics*, Chapter 6

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1. Starting Point: Real Business Cycles (RBC)

What is RBC?

The RBC model combines a stochastic growth model with calibration to generate quantitative implications for business cycles.

Core idea: Economic booms and recessions are *optimal responses* of the economy to technology (productivity) shocks. There is nothing to fix — fluctuations are efficient.

Key assumptions:

- Prices (goods prices and wages) adjust **instantaneously** to clear all markets.
- Markets are perfectly competitive — all agents take prices as given.
- Aggregate supply is **vertical**.

Clarification: What does “aggregate supply” mean?

Aggregate supply (oferta agregada) is simply the total quantity of goods and services that all firms in the economy are willing to produce. “Aggregate” means we sum over all firms in the country.

The Classical Dichotomy

Because aggregate supply is vertical, nominal disturbances (e.g., monetary policy) only affect the *nominal* side of the economy — prices, wages, and money — but not *real* variables such as output and employment.

Clarification: What does “nominal” mean?

Nominal = expressed in money terms (pesos, euros, dollars) — the number on the price tag.

Real = how much you can actually buy with that money.

If the central bank doubles the money supply and aggregate supply is vertical, firms cannot produce more — they can only raise prices proportionally. Real output is unchanged. Money is **neutral**.

Clarification: Is Y always constant in RBC?

No. Y changes, but only due to **real** shocks (technology improvements, labor supply changes). What *is* constant is Y 's response to nominal/monetary disturbances — those cannot move real output in RBC.

Summary: Classical Dichotomy

Aggregate supply is vertical $\Rightarrow$ nominal disturbances only affect the nominal side of the economy. This is the **classical dichotomy**. Money is neutral.

2. Motivation for New Keynesian Models

Incomplete / Sluggish Nominal Adjustment

In reality, prices and wages do **not** adjust instantaneously. This is called **nominal rigidity** (rigidez nominal).

Clarification: What is nominal rigidity?

It means prices or wages cannot easily or quickly adjust. When prices are sticky and demand rises, firms *produce more* rather than simply raising prices. This is the key channel through which monetary disturbances have real effects.

Consequence: When prices are sticky and demand rises, firms *produce more* instead of just raising prices. This breaks the classical dichotomy:

Incomplete nominal adjustment $\Rightarrow$ changes in aggregate demand affect output and employment

$\Rightarrow$ Nominal disturbances cause changes in employment and output (at least in the short run)

Real Shocks and a New Channel

Real shocks (government spending, oil prices, etc.) also affect aggregate demand at a given price level. With incomplete price adjustment, this creates an **additional channel**: these shocks now also affect output and employment more strongly than in RBC, because prices cannot immediately absorb the shock.

Summary: RBC vs. New Keynesian

	RBC	New Keynesian
Prices	Fully flexible	Sticky (rigid)
Money	Neutral	Non-neutral (short run)
What moves Y ?	Only real shocks	Real <i>and</i> nominal shocks

3. A Model of Imperfect Competition and Menu Costs

Setting

Production function of firm i :

$$Y_i = L_i$$

One unit of labor produces one unit of output. Simple.

Household utility:

$$U = C - \frac{1}{\gamma} L^\gamma, \quad \gamma > 1$$

- C : consumption index — more consumption, more utility.

- $-\frac{1}{\gamma}L^\gamma$: *disutility* of working — working hurts, so it has a negative sign.
- $\gamma > 1$: controls how fast the pain of working grows. Higher γ = each extra hour of work costs more in terms of well-being.

Clarification: Why does γ affect how much you work?

From the optimal labor supply result $L = (W/P)^{1/(\gamma-1)}$, a higher γ reduces L because each extra hour of work becomes increasingly painful, so you choose to work less. γ also controls the elasticity of labor supply, $1/(\gamma-1)$: how sensitive your hours are to the real wage.

The CES Consumption Index

$$C = \left[\int_{i=0}^1 C_i^{\frac{\eta-1}{\eta}} di \right]^{\frac{\eta}{\eta-1}}, \quad \eta > 1$$

C_i = how much you consume of good i . The integral sums over all goods in the economy. C aggregates all individual consumptions into one welfare number.

η = **elasticity of substitution**: how easily consumers switch between goods.

- High η : goods are very similar — if a firm raises its price slightly, it loses almost all customers.
- Low η : goods are differentiated — firms have more pricing power.

Simplifications: No capital, no government, no trade $\Rightarrow Y = C$.

Aggregate Demand

$$Y = \frac{M}{P}$$

M = money supply, P = price level, M/P = real money balances.

Three microfoundations for this equation:

1. **Real money balances in the utility function:** Money provides convenience, so it enters utility directly. Optimizing yields $Y = M/P$.
2. **Cash-in-advance constraint:** You must have cash before buying anything: $PC \leq M \Rightarrow Y = M/P$.
3. **Central bank targets nominal GDP:** The central bank sets $PY = M$, directly giving $Y = M/P$.

Clarification: If M rises, doesn't P rise too, leaving Y unchanged?

With flexible prices (RBC): Yes. If M doubles, P doubles, and $Y = M/P$ stays the same. Classical dichotomy holds.

With sticky prices (New Keynesian): P cannot rise immediately. So Y rises in the short run. The key word is *short run* — eventually prices adjust and we return to the classical result. The whole chapter lives in this short run.

What is Imperfect Competition? What are Menu Costs?

Imperfect competition: Firms have market power — they sell a differentiated product and *choose* their own price (unlike perfect competition where the market dictates the price). This is necessary for price rigidity to matter: in perfect competition, prices are already determined by the market, so there is nothing for a firm to “choose” to keep fixed.

Menu costs: The cost of *changing* a price. The name comes from restaurants (literally reprinting the menu). More generally: reprinting catalogues, reprogramming systems, renegotiating contracts. If the cost of changing the price exceeds the gain from changing it, the firm keeps its price fixed. This generates nominal rigidity naturally.

4. The Household's Problem

Stage 1: Allocating the Budget Among Goods

The household has budget S and maximizes consumption subject to spending no more than S :

$$\mathcal{L} = \left[\int_0^1 C_i^{\frac{\eta-1}{\eta}} di \right]^{\frac{\eta}{\eta-1}} + \lambda \left(S - \int_0^1 P_i C_i di \right)$$

- P_i : price of good i .
- C_i : quantity consumed of good i .
- $P_i C_i$: spending on good i .
- $\int_0^1 P_i C_i di$: total spending on all goods = must not exceed S .
- λ : Lagrange multiplier — how much utility rises if you had one more peso of budget.

Clarification: Why does this Lagrangian not include the full utility $U = C - \frac{1}{\gamma} L^\gamma$?

Because the problem is split into two stages. In Stage 1, L is taken as fixed (already decided). Since $-\frac{1}{\gamma} L^\gamma$ is then a constant, maximizing U is identical to maximizing C . The disutility of labor is handled in Stage 2.

Results from solving the Lagrangian:

$$C = \frac{S}{P} \quad (\text{real consumption} = \text{real budget})$$

$$C_i = \left(\frac{P_i}{P} \right)^{-\eta} C \quad (\text{demand for good } i)$$

Clarification: Why the $-\eta$ exponent?

$(P_i/P)^{-\eta} = 1/(P_i/P)^\eta$. So when P_i/P rises (good i becomes relatively expensive), C_i falls. The negative exponent captures the inverse relationship between relative price

and demand. The size of η determines *how strongly* demand responds: high η means consumers are very sensitive to price differences across goods.

The elasticity of demand for each individual good is η .

Stage 2: Choosing How Much to Work

Total income: $S = WL + R$, where WL is labor income and R is profit income (the household owns firms and receives dividends).

Clarification: What is R ?

R is the firm's profit, distributed back to the household as the owner. In this model, households own all firms, so profits return to them as dividend income. Total household income therefore has two components: labor income WL and profit income R .

The household chooses L to maximize:

$$\max_L \frac{WL + R}{P} - \frac{1}{\gamma} L^\gamma$$

FOC for L :

$$\frac{W}{P} - L^{\gamma-1} = 0 \quad \Rightarrow \quad L = \left(\frac{W}{P} \right)^{\frac{1}{\gamma-1}}$$

Intuition: The FOC says: marginal benefit of working = marginal cost of working.
Marginal benefit = W/P = real wage (one extra hour gives you W/P more real income).

Marginal cost = $L^{\gamma-1}$ = marginal disutility of labor (how much extra pain one more hour causes).

What is the real wage W/P ?

W = your nominal salary (in pesos). P = general price level (how expensive everything is). W/P = how much you can *actually buy* with your salary.

If W/P is high $\Rightarrow$ working is worth it $\Rightarrow$ you work more.

If W/P is low $\Rightarrow$ working is not worth it $\Rightarrow$ you work less.

Labor supply is an increasing function of the real wage, with elasticity $\frac{1}{\gamma-1}$.

5. The Firm's Problem

The firm maximizes real profits:

$$\frac{R_i}{P} = \underbrace{\frac{P_i}{P} Y_i}_{\text{real revenue}} - \underbrace{\frac{W}{P} L_i}_{\text{real cost}}$$

Substituting $Y_i = L_i$, the demand function $C_i = (P_i/P)^{-\eta} Y$, and $C_i = Y_i$:

$$\frac{R_i}{P} = \left(\frac{P_i}{P}\right)^{1-\eta} Y - \frac{W}{P} \left(\frac{P_i}{P}\right)^{-\eta} Y$$

FOC for the relative price P_i/P :

$$(1 - \eta) \left(\frac{P_i}{P}\right)^{-\eta} Y + \eta \frac{W}{P} \left(\frac{P_i}{P}\right)^{-\eta-1} Y = 0$$

$$\Rightarrow \boxed{\frac{P_i}{P} = \frac{\eta}{\eta - 1} \cdot \frac{W}{P}}$$

Clarification: What is a markup?

A markup is the factor by which price exceeds marginal cost. In this model, the firm's only cost is the real wage W/P , so the markup is $\frac{\eta}{\eta-1} > 1$.

η	Situation	Markup
2	Few substitutes	$2/1 = 2$
5	Some substitutes	$5/4 = 1.25$
10	Many substitutes	$10/9 \approx 1.11$
$\rightarrow \infty$	Perfect competition	Markup $\rightarrow 1$ (price = marginal cost)

Key result: The price is set as a **markup over marginal cost**. The size of the markup is determined by the elasticity of demand η .

6. Equilibrium

Deriving the Optimal Price in Equilibrium

From the household's FOC: $L = (W/P)^{1/(\gamma-1)}$. In equilibrium, $Y = L$, so:

$$\frac{W}{P} = Y^{\gamma-1}$$

Higher output requires a higher real wage to induce households to supply that much labor.

Substituting into the firm's pricing rule:

$$\frac{P_i^*}{P} = \frac{\eta}{\eta - 1} Y^{\gamma-1}$$

In logarithms (easier to work with):

$$p_i^* - p = \underbrace{\ln \frac{\eta}{\eta - 1}}_c + \underbrace{(\gamma - 1)}_\phi y \equiv c + \phi y$$

The firm's optimal price rises with aggregate output y because higher output means higher wages and thus higher marginal cost.

Symmetric Equilibrium

Since all firms are identical, they all set the same price: $P_i = P$ for all i . This means $P_i^*/P = 1$, so:

$$\frac{\eta}{\eta-1} Y^{\gamma-1} = 1 \quad \Rightarrow \quad Y = \left(\frac{\eta-1}{\eta} \right)^{\frac{1}{\gamma-1}} < 1$$

$Y < 1$ because $(\eta-1)/\eta < 1$. This reflects **monopolistic inefficiency**: firms charge a markup and produce less than the socially optimal level.

Aggregate Demand and the Optimal Price

The aggregate demand equation $Y = M/P$ in logs is:

$$y = m - p$$

Using this to substitute $p = m - y$ into the optimal price equation:

$$p_i^* = c + (1 - \phi)p + \phi m$$

The firm's optimal price depends on:

- p : the general price level. If everything gets more expensive, you raise your price too.
- m : the money supply. More money $\Rightarrow$ more demand $\Rightarrow$ higher optimal price.

7. Connection to New Keynesian Economics

The equilibrium above is the **flexible-price benchmark**: with no adjustment costs, all firms set $P_i = P$ and money is neutral.

The New Keynesian question: What if firms *cannot* adjust their price immediately because of menu costs?

If m rises (central bank prints money), the optimal price p_i^* rises too. But if adjusting costs something and the gain is small, the firm keeps its old price. Then:

- P does not rise proportionally with M .
- $Y = M/P$ rises — more real output.
- Employment rises to meet that output.

Core New Keynesian insight:

Small individual costs of price adjustment (menu costs) can generate large aggregate nominal rigidity. As a result, monetary disturbances affect real output and employment in the short run. The classical dichotomy breaks down.